

NADIM MAHMUD

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PROFESSIONAL SUMMARY

Python Backend Developer with hands-on experience building AI-powered backend services, integrating third-party and internal AI microservices, and delivering scalable production systems. Proficient in Django, Django REST Framework, PostgreSQL, and Docker, with strong experience in asynchronous programming, Celery/Redis background task processing, and WebSocket-based real-time systems. Experienced collaborating with cross-functional engineering teams on complex, multi-role platforms. Strong analytical foundation backed by 600+ competitive programming solutions.

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML, CSS

Frameworks: Django, Django REST Framework, FastAPI (learning), Bootstrap, Tailwind CSS

Databases: PostgreSQL, MySQL, SQLite

Message Brokers & Caching: Redis, RabbitMQ

Backend & Architecture: RESTful API Design, WebSocket, Asynchronous Programming, Celery, Background Task Processing, AI Microservice Integration, System Performance Optimization

Tools & DevOps: Git, GitHub, GitLab, Docker, AWS, VPS, Postman, CI/CD Pipelines

PROFESSIONAL EXPERIENCE

Jr. Backend Developer | Join Venture AI – Dhaka, Bangladesh | Oct 2025 – Present

- Designed, developed, and maintained scalable RESTful backend services using Python and Django REST Framework for AI-powered business applications.
- Integrated third-party APIs and internal AI microservices; implemented JWT-based authentication and role-based access control for secure, production-ready systems.
- Optimized system performance, scalability, and reliability through PostgreSQL query tuning, efficient schema design, and API response optimization.
- Built reliable, non-blocking backend workflows using asynchronous programming and Celery/Redis background task processing, integrated with machine learning pipelines.
- Collaborated with product and engineering teams using Docker and AWS-based cloud deployment workflows, following clean architecture and scalable coding practices.

Team Leader | Vcube Soft and Tech Limited – Dhaka, Bangladesh | Jun 2019 – Oct 2024

- Promoted from Operator to Project Quality Checker, Mentor, and Team Leader through consistent performance.
- Managed team coordination, quality assurance, and workflow optimization for 2D/3D building floor plan projects.
- Mentored new team members and maintained productive communication across remote and in-office teams.

PROJECTS

AI Powered Insurance App

- Built a scalable insurance management platform for a Swedish client featuring a freemium model: free-tier users interact with a real-time AI chatbot via WebSocket for insurance guidance, while premium subscribers access full case management with document submission and AI-driven, document-aware guidance.

- Managed WebSocket session flow and routed structured conversation data to an internal AI microservice for context-aware chat responses; handled all connection management and session-based conversation state.
- Implemented manual email-based registration, Firebase-powered social authentication (Google, Apple), in-app purchase subscription system, role-based access control, JWT-based authentication, and Celery/Redis background tasks for automated weekly insurance claim reminder notifications.

Eat At Home

- Developed a meal tracking and nutrition analytics platform for a US client; users log daily meals and the system generates detailed analytics and monthly reports comparing home versus restaurant spending and nutritional patterns.
- Called an internal AI microservice endpoint to analyze food photos and return restaurant pricing estimates; integrated results into weekly meal planning logic and Celery-based automated scheduling.
- Integrated Plaid API to fetch and analyze users' real bank transaction history, enabling accurate restaurant spending tracking and automated categorization of food-related expenses.
- Implemented manual email-based registration alongside Firebase-powered social authentication (Google, Facebook, Apple); built scalable backend using Django ORM, class-based views, serializers, and routers, with PostgreSQL for analytics data storage and query optimization.

RnD Taxbot

- Built a tax rebate automation platform for an Australian client that guides new organizations through an AI-powered interview process to generate R&D tax rebate applications, with a multi-role approval workflow: user submission → admin-triggered AI scoring → approve or reject with resubmission flow.
- Orchestrated multi-turn AI interview flow via WebSocket, passing structured session data to an internal AI microservice for automated R&D scoring report generation.
- Designed secure document generation and access control so approved reports are visible to both users and admins, while rejected submissions prompt users to update and resubmit; optimized PostgreSQL schema for complex multi-stage approval state management.

Animal Service

- Built a full-featured pet e-commerce and services platform supporting a buyer/seller marketplace for pets, feed, and accessories; integrated auction system, transport booking, and veterinary appointment scheduling (online and offline) with prescription generation and medical history/vaccination record management.
- Implemented a digital wallet system enabling sellers to receive earnings from marketplace transactions and withdraw funds, with automated balance management and transaction history tracking.
- Integrated Stripe payment gateway; implemented real-time 1-on-1 buyer-seller chat via WebSocket and a comprehensive admin panel with full analytics, fraud prevention controls, and centralized management of transport provider and veterinary doctor verification workflows.
- Implemented multi-method registration supporting both email and phone number sign-up; designed role-based access control for buyers, sellers, admins, veterinary doctors, and transport providers; optimized PostgreSQL schema and queries across marketplace, bookings, medical records, and transactional workflows.

EDUCATION

Bangladesh Sweden Polytechnic Institute

Diploma in Computer Technology | Rangamati, Bangladesh | Jan 2014 – Jun 2018

KEY ACHIEVEMENTS

- Solved 600+ programming problems across Codeforces, LeetCode, HackerRank, CodeChef, and InterviewBit.
- Secured 38th place in the weekly CodeChef contest by solving all problems in June 2025.
- Achieved 11th position in the Inter Polytechnic Programming Contest at IUBAT.